ERIGA ROBERT

SEDev Masters Student | Solar PV & Electrical Engineer | Renewable Energy Analyst & Audit | AI & ML

Flensburg, Germany | +4915907914616

erigarobert2@gmail.com | linkedin.com/in/eriga-robert-9522b518a

PROFESSIONAL PROFILE

Solar PV and electrical engineer with 3+ years of experience designing, bidding, commissioning and operating solar PV, hybrid and grid-connected systems, with 161 kV and 33 kV substation exposure. Contributed to winning tenders, including projects under UNDP in Sierra Leone, and presented designs to clients. Skilled technical writer and analyst, applying MATLAB, Simulink and Python to machine-learning research. Gold medalist in Electrical Engineering; M.Eng. candidate in Sustainable Energy and Development (SEDev).

TECHNICAL SKILLS

Solar PV engineering: design and sizing, off-grid, hybrid and grid-connected plants, battery storage, testing and commissioning.

Substation, protection and studies: 161 kV switchyard, 33 kV collection, GIS, transformers up to 20 MVA, switchgear, protection coordination, load flow and short-circuit studies, SCADA/PLC.

Energy analysis and audit: daily generation analysis, plant performance and availability assessment, loss and fault identification.

Technical writing: proposals and tender documents, design reports, specifications, bills of quantities, performance reports, research papers.

AI and machine learning: model development and simulation in MATLAB, Simulink and Python, LLM knowledge.

WORK EXPERIENCE

Electrical Engineer & Energy Analyst | **Frontier Energy – Planet Solar Energy** **Nov 2025 – Jul 2026**

Freetown, Sierra Leone

- **Sustained plant availability:** executed electrical operations and maintenance across a 50 MW grid-connected solar PV portfolio (35 MW operational, 15 MW under implementation).
- **Drove performance through data:** analysed daily generation against expected output and isolated inverter, transformer and feeder losses, enabling faster corrective action; authored performance reports for management.
- **Mastered the grid interface:** worked across 1,500 V DC generation, 33 kV collection and the 161 kV connection to the Bumbuna–Freetown corridor.
- **Resolved faults fast:** inspected and troubleshot circuit breakers, CTs, CVTs/VTs, surge arresters, protection relays and transformers, using SCADA/PLC alarms to cut avoidable outages; supported pre-energisation checks and commissioning of new capacity.

Tools: *SCADA/PLC, ABB Automation, Movicon, Microsoft Excel, PV analyzer, Electrical Testing, Python.*

Electrical Engineer – Solar PV Design, Tenders & Implementation | **Aptech Africa Limited** **Aug 2023 – Sep 2025**

Kampala, Uganda

- **Won work through bids:** wrote technical proposals, designs and tender documents that contributed to winning bids for multiple solar projects(community and government laboratories), including projects under UNDP in Sierra Leone.
- **Presented designs:** presented system designs to clients and stakeholders, translating energy needs into PV, battery and hybrid configurations.
- **Healthcare and utility-scale solar:** contributed to a 2.4 MW hybrid PV project for seven hospitals with 100 kWh storage, and supported a 12 MW plant near Juba, South Sudan, and its 26 MW expansion and also upgrade of the five government hospital laboratories with Battery Banks and Battery Management System (BMS)
- **Design and delivery:** sized arrays, batteries and inverters for off-grid and hybrid systems; carried out installation checks, testing and commissioning.

Tools: *PVsyst, HOMER ,Etap, Matlab, AutoCAD, Microsoft Office*

Electrical Engineering Project Experience | **Tororo Cement Limited** **Project-based**

Tororo, Uganda

- Participated in GIS and 20 MVA indoor transformer installation and in commissioning 11 switchgear panels; contributed to protection coordination; assisted with a 5,800 kW motor for VRM Project II under ABB project.

EDUCATION

M.Eng. Sustainable Energy and Development | Europa-Universität Flensburg, Germany **Sep 2026 – Present**
Energy Analysis, Energy-system modelling, renewable energy and sustainable development.

B.Tech. Electrical Engineering | CV Raman Global University, India **2019 – 2023**
CGPA 8.31/10; Gold Medal for Academic Excellence.

ENGINEERING PROJECTS

2.4 MW Hybrid Solar PV for Seven Hospitals | Aptech Africa

- **Applied:** sized and designed a PV-battery-diesel system with 100 kWh storage to match hospital loads.
- **Impact:** dependable electricity for essential healthcare services with Battery Management System.

12 MW Solar Plant and 26 MW Expansion, Juba, South Sudan | Aptech Africa

- **Applied:** supported electrical engineering, testing and commissioning of a grid-connected plant and its expansion.
- **Impact:** added grid-connected renewable capacity.

Solar Project Bids, Sierra Leone | Aptech Africa

- **Applied:** wrote technical proposals, designs and supporting documents, and presented the designs to clients.
- **Impact:** contributed to winning several solar project bids.

Machine-Learning Research for Energy Systems | MATLAB, Simulink, Python and Large Language Model (LLM).

- **Applied:** developed and simulated machine-learning models for energy and power-electronics applications, combining Simulink system models with Python data analysis and now working on O & M projects using LLM.
- **Impact:** a data-driven approach to system performance and control, documented in technical research reports.

Solar-Powered EV DC Fast Charging and Converter Control | IEEE-indexed publication

- **Applied:** modelled and controlled the charging converter and documented the results as a technical paper.

Sustaina Nexus Foundation | (Founder) - Initiative in development

- **Applied:** designing an energy-access and agriculture initiative for underserved rural communities in Uganda through community mobilization and grant funding.

Energy-System Modelling and Power-Plant Efficiency Analysis | M.Eng. coursework

- **Applied:** reviewed energy-system modelling tools and analyzed fuel-to-electricity efficiency of power plants across countries and years.

CERTIFICATIONS AND TRAINING

- Totally Integrated Automation (CV Raman Siemens Centre of Excellence); Industrial Automation (CV Raman Bosch Rexroth Centre of Excellence).
- Mitacs Globalink Research Internship, University of Guelph, Canada (2022);
- Project Management, Coursera/Google (2024).

SOFTWARE AND TECHNICAL TOOLS

Solar and power systems: PVsyst, HOMER, ETAP, DIgSILENT Power Factory, PSCAD.

Design and computing: AutoCAD, MATLAB/Simulink, Python, ABB Automation.

Monitoring and office: SCADA/PLC tools, Microsoft Word, Excel and PowerPoint.

PERSONAL SKILLS

Technical communication: turns complex designs into clear proposals and reports.

Analytical problem solving: traces faults and performance losses to root cause using data.

Initiative and leadership: founded Sustaina Nexus Foundation and is developing.

Cross-cultural teamwork: has worked and studied in India, Canada, Uganda, South Sudan, Sierra Leone and Germany.

Reliability and attention to detail: careful, safety-minded and dependable when working on live high-voltage equipment and deadlines.

LANGUAGES AND REFERENCES

English: professional working proficiency | **German:** A2

References: available on request